

SHETH L.U.J. AND SIR M.V. COLLEGE

12 Generating correlation matrices using cor() (R).

The screenshot shows the RStudio interface with the following content:

```
> quality_data <- read.csv("updated_quality_of_life_2024.csv")
> head(quality_data)
  Country Year Quality_of_Life_Index Purchasing_Power_Index Safety_Index
1 Norway 2024 182.3 88.4 75.1
2 Switzerland 2024 179.5 122.3 70.8
3 Australia 2024 176.8 108.6 60.5
4 Germany 2024 170.4 112.9 64.3
5 Canada 2024 172.1 105.4 63.2
6 Japan 2024 168.9 96.2 77.5
  Health_Care_Index Cost_of_Living_Index Pollution_Index
1 75.6 75.0 21.2
2 71.2 122.4 19.4
3 73.4 73.9 23.5
4 74.1 65.3 28.7
5 72.9 67.1 26.4
6 80.1 82.1 38.9

> colnames(quality_data)
[1] "Country" "Year" "Quality_of_Life_Index"
[4] "Purchasing_Power_Index" "Safety_Index" "Health_Care_Index"
[7] "Cost_of_Living_Index" "Pollution_Index"
> numeric_data <- quality_data[sapply(quality_data, is.numeric)]
> correlation_matrix <- cor(numeric_data, use = "complete.obs")

Warning message:
In cor(numeric_data, use = "complete.obs") : the standard deviation is zero

> print(correlation_matrix)
      Year Quality_of_Life_Index Purchasing_Power_Index Safety_Index
Year 1 NA NA NA NA
Quality_of_Life_Index 1.0000000 0.9242566 0.7246225 0.7246225
Purchasing_Power_Index NA 0.9242566 1.0000000 0.5249637
Safety_Index NA 0.7246225 0.5249637 1.0000000
Health_Care_Index NA 0.9482368 0.8497637 0.8205545
Cost_of_Living_Index NA 0.8404467 0.8463137 0.7035896
Pollution_Index NA -0.9841527 -0.9113347 -0.6737738
      Health_Care_Index Cost_of_Living_Index Pollution_Index
Year NA NA NA
Quality_of_Life_Index 0.9482368 0.8404467 -0.9841527
Purchasing_Power_Index 0.8497637 0.8463137 -0.9113347
```

The screenshot shows the RStudio interface with the following content:

```
> packages_data <- read.csv("updated_packages_2024.csv")
> head(packages_data)
  Package Year Downloads_12_Months Rating
1 R 2024 1000000 4.5
2 Python 2024 500000 4.0
3 Java 2024 200000 3.5
4 JavaScript 2024 300000 4.2
5 C++ 2024 150000 3.8
6 PHP 2024 100000 3.0
  Health_Care_Index Cost_of_Living_Index Pollution_Index
1 75.6 75.0 21.2
2 71.2 122.4 19.4
3 73.4 73.9 23.5
4 74.1 65.3 28.7
5 72.9 67.1 26.4
6 80.1 82.1 38.9

> colnames(packages_data)
[1] "Package" "Year" "Downloads_12_Months" "Rating"
[4] "Health_Care_Index" "Cost_of_Living_Index" "Pollution_Index"
> numeric_data <- packages_data[sapply(packages_data, is.numeric)]
> correlation_matrix <- cor(numeric_data, use = "complete.obs")

Warning message:
In cor(numeric_data, use = "complete.obs") : the standard deviation is zero

> print(correlation_matrix)
      Year Downloads_12_Months Rating
Year 1 NA NA NA
Downloads_12_Months 1.0000000 0.9242566 0.7246225
Rating NA 0.9242566 0.5249637 1.0000000
Health_Care_Index NA 0.9482368 0.8497637 0.8205545
Cost_of_Living_Index NA 0.8404467 0.8463137 0.7035896
Pollution_Index NA -0.9841527 -0.9113347 -0.6737738
      Health_Care_Index Cost_of_Living_Index Pollution_Index
Year NA NA NA
Downloads_12_Months 0.8404467 0.8463137 -0.9841527
Rating 0.8497637 0.8463137 -0.9113347
```

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ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R